Blackjack Probability

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The problem

- The dealer draws cards valued 1-10 uniformly while below target-4; busts if above target.
- Starting from a hand value, probability of busting (rounded to 3 decimals).
- Memoised recursion.

Intuition

- The chance of busting from hand total t is the average over the ten next cards, which we memoise.

Brute force vs optimal

- Plain recursion re-explores the same hands → exponential
- Memoise by hand value → O(target) states × 10 cards
- Bottom-up DP works equally well

Approach

- 1. If the total exceeds the target, it's a bust
- 2. If it reaches target minus four, stop
- 3. Otherwise average the ten draws
- 4. Memoise by total

Worked example

P(bust) for hands 15...21



$$P(\text{bust} \mid 16) = 0.1 \times \Sigma\, P(\text{bust} \mid 16 + \text{card}) = 0.500$$

Hand 16: P(bust) = 0.500

Worked example

P(bust) for hands 15...21



$$P(\text{bust} \mid 15) = 0.1 \times \Sigma\, P(\text{bust} \mid 15 + \text{card}) = 0.450$$

Hand 15: P(bust) = 0.450

Java solution

```java
public class BlackjackProbability {
    public static float blackjackProbability(int target, int startingHand) {
        double p = bust(target, startingHand, new HashMap<>());
        return Math.round(p * 1000) / 1000f;
    }

    private static double bust(int target, int hand, Map<Integer, Double> memo) {
        if (hand > target) return 1;
        if (hand + 4 >= target) return 0;
        Double m = memo.get(hand);
        if (m != null) return m;
        double p = 0;
        for (int c = 1; c <= 10; c++) p += 0.1 * bust(target, hand + c, memo);
        memo.put(hand, p);
        return p;
    }
}
```

Complexity

- Time: O(t)
- Space: O(t)

Edge cases & pitfalls

- Stand when hand ≥ target − 4
- Each card 1–10 has probability 0.1
- Round to 3 decimals at the end only

Interview tips & follow-ups

- Explain the state: just the hand total
- Follow-up: realistic deck (cards without replacement)

Recap

- The chance of busting from hand total t is the average over the ten next cards, which we memoise.
- Complexity: O(t)